

Q1)

1. F, Filter Shape (σ)
2. T
3. F
4. T
5. T
6. T
7. T
8. T
9. T
10. T
11. F
12. F ,

1	2	1
0	0	0
-1	-2	-1

13. T
14. F, **Faster**
15. F, Window Size
16. T
17. T
18. F, False Positive
19. T
20. T

Q2)

A)

Filter:

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Operation: Sharpening

B)

- I. Laplacian of Gaussian, it's used for edge detection in images
- II. The zeros correspond to Edges (zero-crossing)
- III. It represents the shape of the gaussian filter, a higher value would increase the smoothing thus decreasing the number of detected zeros (edges)
- IV. B, Since performing the derivative on the gaussian filter is more efficient than performing it on the whole image after convolving it with the filter

C)

- I. Pair 1, Since the harris detector is rotation **invariant** but scale variant.
- II.

$$\begin{bmatrix} 29 & 97 \\ 97 & 353 \end{bmatrix}, \text{ The values correspond to an } \mathbf{Edge} \text{ because only } I_y \text{ is large}$$

D)

- Iterate through each point and compare to its 8 neighbors in the current image and 9 neighbors each in the scales above and below.
- A point is marked as key point if it is greater or smaller than all 26 neighbors.

↑

lec 5, slide 13

Max points are **8** in **scale 4**

- I. Location 3,2 or 2,3
- II. Scale : 4

Q3)

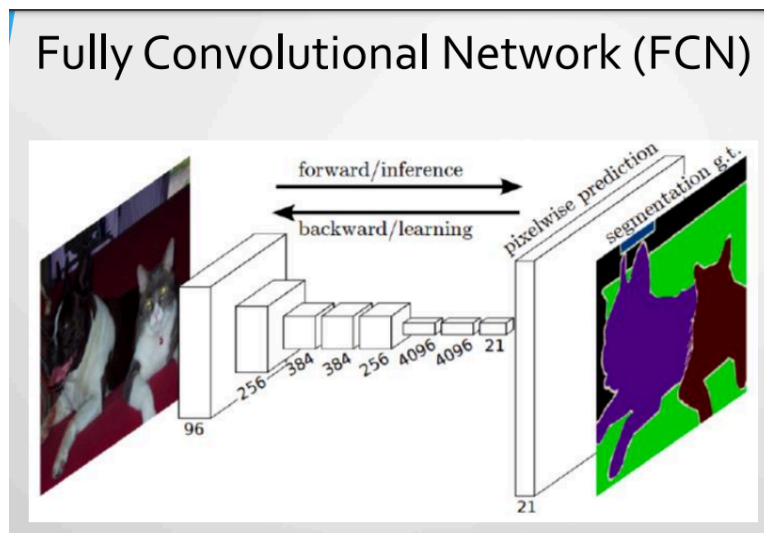
A)

- 1) From all the photos (including test photo):
 - a) Extract the features of the photo using SIFT
 - b) Learn “visual vocabulary”
 - c) Quantize features using visual vocabulary
 - d) Represent photos by frequencies of “visual words”
 - e) Convert each histogram into a normalized feature vector
- 2) For each photo in the training set
 - a) Calculate distance/similarity between Train image and test image using Euclidean Distance or Cosine Similarity
 - b) Output the similarity/distance score

B)

1. ConvNet, It's used to extract features from the image
2. Selective Search
3. ROI pooling, used to resize/wrap the region proposals to a fixed size
4. A vector of probabilities for each class (including background), the classifier used is **Softmax**
5. The attributes of the bounding box

C)



The skip connections improve the performance by passing the output of the pooling layers to the upsampling layers